

Perspective

Clinical connectomics: Mapping structural brain networks in disease

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SUMMARY

The human connectome holds considerable promise for advancing an integrated, systems-level understanding of brain disorders. Critical to this ambition is accurate, reliable, and biologically informative connectome mapping and analysis in the presence of diverse brain pathologies. However, many connectome-mapping methodologies have been established and validated using healthy, non-clinical participants. Here, we consider the challenges of structural connectome mapping in the presence of pathology. We first classify connectome abnormalities and relate them to underlying white-matter microstructure. We explain how measures of structural connectivity can quantify the impact of microstructural white-matter pathology on the macro-scale connectome, including measures based on streamline counts, microstructure-informed tractography, tract-averaged metrics, and structural similarity. We consider connectome-mapping pipelines and algorithmic choices to facilitate reliable and biologically meaningful clinical connectome mapping. Our aim is to establish best practices for connectome mapping and analysis in neurology, psychiatry, and neurosurgery.

INTRODUCTION

Over the last two decades, connectome mapping using diffusion magnetic resonance imaging (MRI) and tractography has evolved into a mature technique for studying brain circuits and network organization *in vivo*.^{1–8} While several tractography limitations and biases have surfaced during this period,^{9–11} ongoing progress in human connectomics is paving the way for groundbreaking clinical applications. Termed clinical connectomics, these applications encompass preclinical research as well as clinical translation of connectome techniques to understand, diagnose, and optimize therapeutics for neurological and psychiatric illnesses.

Soon after the first human connectome was mapped,^{12,13} connectome abnormalities were uncovered in schizophrenia¹⁴; followed by the discovery of similar abnormalities in numerous other brain disorders.^{15–19} Initially, research focused on modeling the connectome as a graph and testing for patient-control differences in graph topology. Schizophrenia and many other psychiatric illnesses were linked with altered topological measures, including small-worldness, hub structure, rich-club architecture, and modular organization.^{20–24} Indeed, there are now few major neurological and psychiatric disorders for which brain network measures have not been characterized. Patient-control differences in connectome topology provide profound insight into abnormal brain network organization,^{25,26} but these topological discoveries have not led to tangible clinical translation.

In the post-small-world connectome era, groundbreaking clinical applications are emerging, some of which are now in clinical use (Figure 1). For example, connectomes and tractography are

used to identify optimal targets for focal brain stimulation therapies^{27,28} (see Box 1). Deep brain electrode stimulation targets can be personalized to modulate disease- or symptom-specific distributed brain circuits, which, when modulated, yield maximal therapeutic response.²⁹ Although functional connectivity is more commonly used to optimize connectivity-based targets for transcranial magnetic stimulation (TMS) therapy,^{30–32} connectome analyses can reveal the anatomical pathways along which stimulation effects propagate to distant functionally-defined targets.³³ Beyond treatment target identification and neurosurgical planning,^{34–36} tractography-derived connectomes can serve as putative biomarkers for diagnostic and prognostic prediction models,³⁷ monitoring disease progression³⁸ and disease subtyping.³⁹

The success of clinical connectomics hinges on accurate, reliable, and biologically informative connectome mapping in pathology. But most connectome-mapping methodologies are tested and validated in the absence of pathology, leaving the method's potential to reliably characterize connectome abnormalities underexplored. Mapping connectomes in disease gives rise to unique challenges, many of which remain poorly understood in the field.

Furthermore, the clinical field still relies on arguably outdated tractography algorithms and connectivity measures. The criticized streamline count connectivity measure⁴³ and tensor-based tractography algorithms⁴⁴ remain entrenched in clinical research and translation, despite recognized limitations. Facilitating the transition to state-of-the-art connectome-mapping tools in the clinical community is needed, but only if those tools can reliably and accurately deal with disease-related brain



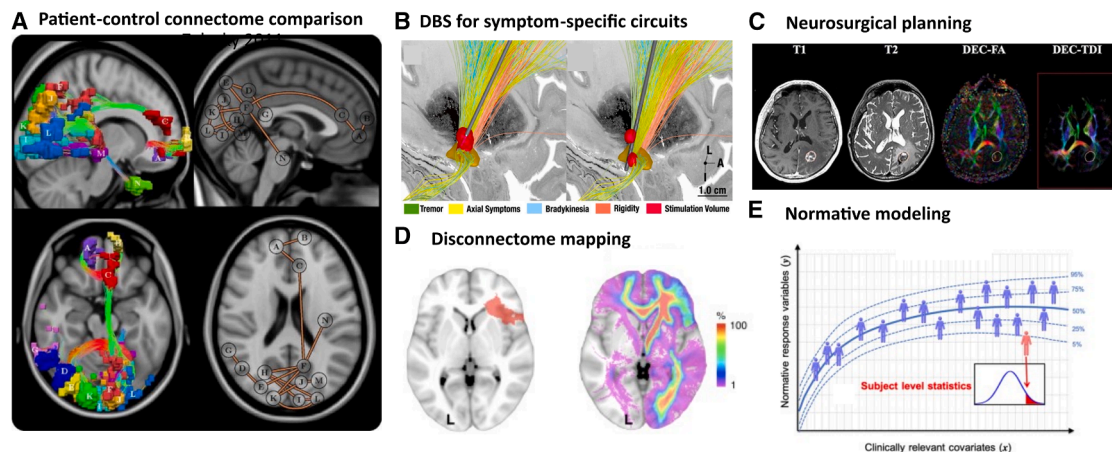


Figure 1. Current and future clinical applications of human connectomics and diffusion MRI tractography

(A) Altered white-matter connectivity comprising the schizophrenia connectome identified using patient-control comparison. Reproduced from Zalesky et al.²⁴ with permission.

(B) Targeting symptom-specific circuits with deep brain stimulation (DBS): A standard omnidirectional electrode covers all tracts (left), whereas a modern 15-contact electrode (right) enables steering stimulation toward individual tractography-derived tracts linked with specific symptoms of Parkinson's disease. Reproduced from Rajamani et al.²⁹ under Creative Commons CC BY 4.0.

(C) The directionally encoded (DEC) track density image (TDI, far right) is derived from diffusion MRI tractography and provides tract-level spatial information to potentially support decision-making around tumor resection extent, compared with the fractional anisotropy (FA) and structural images. Reproduced from Kamagata and Andica³⁴ with permission.

(D) Disconnectome mapping: A brain lesion (left) and the network of disconnections (right) resulting from it can yield a better prediction of dysfunction and recovery from stroke than the lesion itself. Reproduced from Thiebaut de Schotten et al.⁴⁰ under Creative Commons CC BY 4.0.

(E) The normative reference chart shows variation in a brain phenotype, such as structural connectivity, as a function of age or other relevant variables. Reproduced from Marquand et al.⁴¹ under Creative Commons CC BY 4.0.

pathology. Moreover, given the higher technical entry barriers to structural connectome mapping compared with functional connectivity studies, adequate training and guidance are essential.

This perspective addresses the challenges of building and analyzing tractography-derived connectomes in disease, aiming to inform clinical researchers and clinicians alike on best practices in the field. Our focus is on the methodological foundations of clinical connectome mapping, emphasizing concepts and principles over specific tools and software. We do not consider functional brain networks and refer the reader to earlier reviews on this topic.^{25,45,46} Although tractography is a key component of connectome mapping, our focus is not on tractography in pathology *per se*, as this has been comprehensively addressed in prior work.^{36,47} We begin by defining what constitutes a connectome abnormality and consider the challenges in attributing specific white-matter pathologies to disease-related connectome changes, including the unresolved microstructural substrates of these macroscale changes. We next provide recommendations for mapping and analyzing clinical connectomes, focusing on the choice of connectivity measures, tractography pipeline, and other algorithmic choices. Our intention is not to design new disease-specific connectome-mapping pipelines, but rather to provide recommendations, methodological guidance, and advice to enable reliable and accurate mapping of connectomes in disease. Finally, we briefly consider the prospects for clinical connectomics and present strategic directions for future progress. Overall, this perspective aims to provide guidelines to facilitate best practices in mapping and analyzing clinical connectomes. We assume readers have a basic familiarity with connectome-mapping methods, their terminology, and challenges.

WHAT IS A CONNECTOME ABNORMALITY?

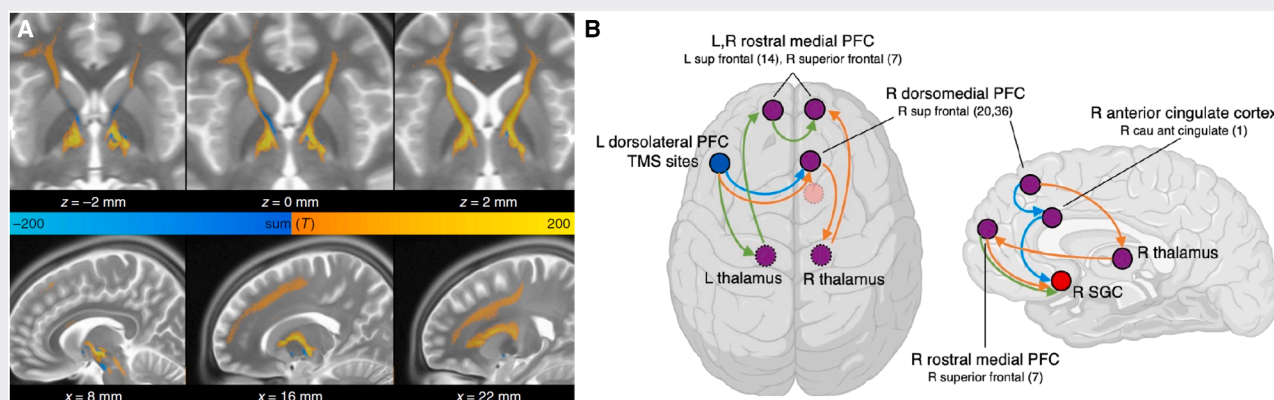
Abnormalities in the connectome can take many forms. They are the results of pathological or developmental changes in white-matter microstructure (Figure 2A), which in turn manifest macroscale topological disruptions within a brain graph model⁴⁸ (Figure 2B, right), or changes in the strength of white-matter connectivity⁴⁹ (Figure 2B, left). Connectome abnormalities can span different spatial scales, from local effects circumscribed to distinct nodes, connections, or circuits, to global changes in network organization.⁵⁰ For example, in stroke, focal epilepsy and brain tumors, abnormalities are localized to specific nodes and white-matter tracts (but network-level effects may emerge elsewhere in the connectome),⁵¹ whereas schizophrenia and many psychiatric disorders are characterized by diffuse, subtle and widespread abnormalities, which may be more difficult to localize.²³

Apart from rare disorders such as agenesis of the corpus callosum, brain tumors, and severe brain malformations or surgical interventions,⁵² connections are rarely absent. Rather, connectome abnormalities are developmental or disease-related changes in the strength of interregional axonal connectivity. These interregional changes in connectivity strength can give rise to disruptions in network organization and topology.

Connectome abnormalities are commonly studied as group-level effect maps derived from statistical inference, individual deviations from normative models, or features of a predictive model (Figure 2C). They are most commonly inferred using patient-control comparisons,^{53,54} lesion network mapping⁵⁵ and related disconnectome approaches.⁴⁰ A connectivity matrix, network visualization, or brain rendering can then be used to

Box 1. Using structural connectivity to optimize therapeutic brain stimulation

An expanding body of work highlights the potential of diffusion MRI tractography-based structural connectivity to optimize deep brain stimulation (DBS) and transcranial magnetic stimulation (TMS) target selection. In obsessive compulsive disorder, the extent to which the subcortical DBS target is structurally connected via a white-matter tract to the ventrolateral prefrontal cortex and dorsal anterior cingulate cortex predicts treatment response (A).²⁸ This work unifies heterogeneity in therapeutic response across subcortical stimulation targets as being driven by whether DBS successfully engages the intended tract. Li and colleagues identified this tract by systematically estimating connectivity between stimulation sites and all other brain regions across individuals, and then testing whether interindividual variability in streamline counts to each brain region—arising from differences in stimulation location—was associated with clinical outcomes.⁴² In major depressive disorder, TMS is typically delivered to the dorsolateral prefrontal cortex (DLPFC), with the aim of modulating activity in the subgenual anterior cingulate cortex (sgACC). There is limited evidence for a direct white-matter tract connecting these two structures; instead, structural connectome analyses suggest a polysynaptic pathway in which stimulation effects may propagate through multiple intermediate regions, including the thalamus and anterior cingulate cortex, before reaching the sgACC (B).³³ The efficiency of the polysynaptic pathways depends on the precise DLPFC stimulation site, with Seguin and colleagues showing that clinical response is related to the structural efficiency of these pathways. These findings are important for stimulation target selection during DBS and TMS treatment planning, as they indicate that a target's structural connectivity profile can potentially predict individual treatment response.



(A) Brain rendering of white-matter tract associated with DBS efficacy in obsessive compulsive disorder. Reproduced from Li et al.²⁸ under Creative Commons CC BY 4.0. (B) Network representation of polysynaptic pathways from DLPFC stimulation site to sgACC target associated with TMS efficacy in depression. Reproduced from Seguin et al.³³ under Creative Commons CC BY 4.0.

encode/visualize loci of abnormal connections, nodes, or topological properties, in which effect sizes or the Bayes factor are used to quantify the magnitude of between-group differences and strength of evidence, respectively.

Group-level characterizations do not capture interindividual heterogeneity in the pathological loci and severity. Heterogeneity is especially evident in psychiatric disorders, in which group-level patient-control differences are often not representative of any given patient.^{56–58} Individual-specific inference can provide a more meaningful characterization in these circumstances and enables connectome-based prognostics and personalization of therapeutics. However, individual inference is challenging because most current measures of tractography-derived connectivity do not provide absolute quantification, making it difficult to determine whether a connectivity measurement is unusually high or low. Normative modeling and AI-based approaches will likely enable the identification of individual abnormalities in the future by establishing normative reference ranges conditioned on an individual's age, sex, and other factors.^{59,60} Lifespan normative charts for functional connectomes⁶¹ and structural similarity networks⁶² are now available to quantify individual deviations from age- and sex-specific normative ranges of connectivity

strength, enabling investigation of pathology at the individual level. Normative charts for tractography-derived measures of connectivity remain to be developed.

In the remainder of this section, we consider how and where abnormalities can manifest in the connectome.

Pairwise connectivity abnormalities

Pairwise connections (i.e., edges) are the fundamental building blocks of the connectome.⁶³ A connection is defined by one (or potentially multiple) white-matter fiber tracts interconnecting a pair of gray-matter regions. Connections have a connectivity strength, referred to as the *interregional connectivity strength* or *weight*. The choice of connectivity strength measure is vital, and we consider various connectivity measures below (see [choosing a connectivity measure](#)). Although often expressed as higher-order network and topological properties (see below), between-group differences in measures such as small-worldness, modular organization, hub structure, and other topological properties ultimately arise from between-group differences in pairwise connections. Connectivity strength differences can be isolated to a specific connection, but more commonly, manifest across multiple circuits and subnetworks.

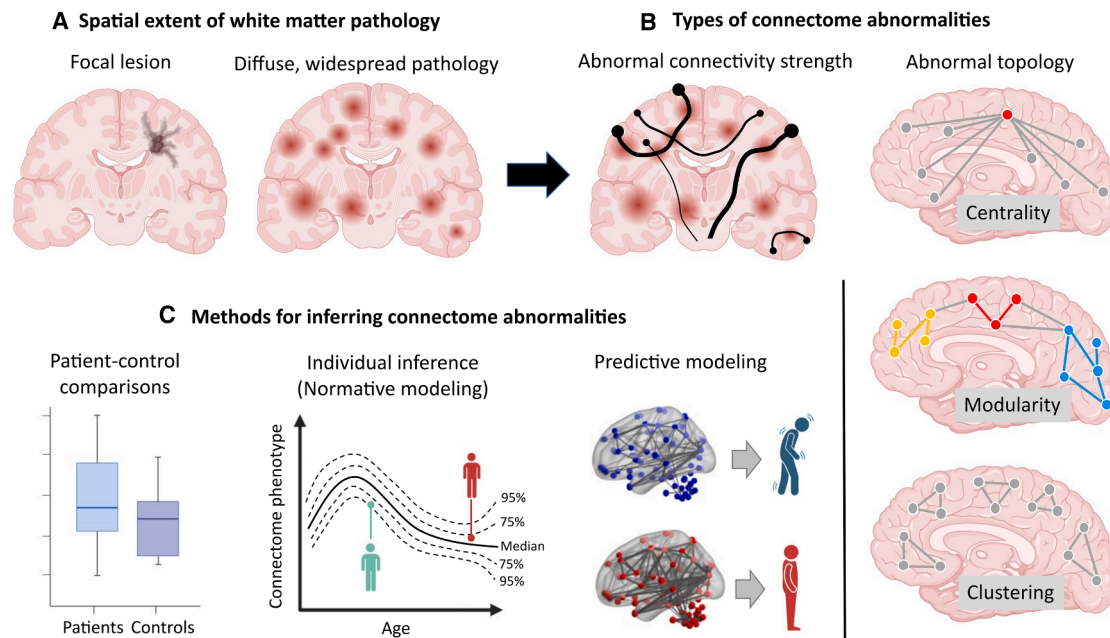


Figure 2. Characterizing abnormalities in the human connectome

(A) Coronal brain images distinguish focal and spatially circumscribed white-matter pathology, such as Multiple Sclerosis lesions (left), versus diffuse, widespread, and non-localizable pathology, such as diffuse axonal injury (right). Microstructural white-matter pathology can manifest as abnormalities of the macroscale connectome, indicated by the arrow.

(B) Connectome abnormalities can be studied as disruptions in interregional connectivity strength (left) or abnormalities in network organization, known as topology (right).

(C) Each image represents a method for inferring connectome abnormalities. Patient-control comparison (left) is the most common paradigm, but does not enable patient-specific inference afforded by normative modeling (center). Predictive models (right) use connectome features to predict disease onset, clinical subtypes, and treatment response likelihood.

Topological abnormalities

The number of connections comprising the connectome can range from thousands to tens of thousands. Studying abnormalities at the resolution of individual connections can become intractable and potentially obscure network-wide organizational changes. Graph-theoretic measures provide a means to understand the collective impact of widespread connection-level abnormalities on overall network organization.^{48,50} A vast array of graph-theoretic measures and properties can be compared between patients and control groups to elucidate topological abnormalities.⁵⁴ Some of these measures enable localization of topological abnormalities to specific nodes (e.g., hubness measures)⁶⁴ and connections (e.g., edge betweenness), whereas other measures quantify network-wide abnormalities (e.g., small-world index).⁶⁵ Graph-theoretic studies in clinical connectomics aim to utilize a carefully selected subset of these measures to understand how disease and divergent brain development alter the network organizational structure of the connectome, known as its topology. Highly exploratory studies evaluating a vast number of graph-theoretic measures are subject to growing criticism due to concerns about the interrelatedness of network measures,⁶⁶ multiple-comparison correction, and the difficulty in relating patient-control differences in abstract network properties to concrete pathological processes.

Connectome coupling abnormalities

The connectome is tightly coupled to the function and structure of many body systems. For example, the connectome constrains

functional connectivity^{67,68} and individual differences in connectome properties have been linked to variation in other physiological, genetic, and metabolic systems.^{69,70} In disease, normal coupling between the connectome and other brain and body systems can be disturbed.⁷¹ Most research to date has focused on understanding altered coupling between the connectome and functional brain networks inferred from resting-state functional MRI.^{72–74} Functional connectivity may provide greater sensitivity to connectome pathology than tractography-derived connectivity measures, particularly when examining longitudinal connectivity changes over a relatively short time period. For example, acute therapeutic effects, such as those induced by brain stimulation, may lead to immediate changes in functional connectivity and metabolic measures (e.g., fMRI and positron emission tomography [PET]),⁷⁵ whereas corresponding macroscale structural remodeling is likely to occur over substantially longer periods, if at all.

WHAT MICROSTRUCTURAL PROCESSES UNDERLIE CONNECTOME ABNORMALITIES?

Connectome abnormalities studied in humans using tractography and diffusion MRI are macroscale, distinguishable at scales no finer than approximately 1–2 mm. The pathological and developmental white-matter processes that underlie connectome abnormalities involve axons, synapses, and neurons, which cannot be resolved at this resolution with diffusion MRI. The diffusion

signal reflects the ensemble behavior of water molecules interacting with thousands to millions of axons. Unfortunately, an orders-of-magnitude gap in resolution exists between macroscale connectome abnormalities and the axonal pathologies that are hypothesized to underlie them.

This mismatch in resolution between tractography and histology presents a major challenge toward linking white-matter microstructural pathologies with the connectome. Elucidating this link is critical to allow the field to draw more nuanced and specific conclusions about patient-control connectome differences. Progress has been made to this end. Early rodent studies linked changes in certain microstructural white-matter measures derived from diffusion MRI to specific axonal processes.⁷⁶ For example, changes in radial diffusivity signify myelin sheath damage, whereas lower axial diffusivity reflects axonal degeneration or loss of axonal integrity.^{77–79} Quantifying connectivity strength with tract-averaged radial or axial diffusivity may thus provide some insight into myelin or degenerative pathologies. Connectivity strength can also be quantified using free water imaging⁸⁰ and neurite orientation dispersion and density imaging (NODDI)⁸¹ to distinguish putative white-matter inflammation and neurite packing density, respectively.^{82,83} While offering greater microstructural specificity, these advanced measures are not pathologically definitive. For example, changes in neurite density index inferred from NODDI can indicate healthy white-matter maturation in children⁸⁴ but also gliosis.

Multiscale studies are needed to bridge connectomics and cellular resolutions and reveal the pathological processes that underlie a connectome abnormality. This may involve a range of strategies, including multimodal imaging, postmortem histology, and animal models. One promising strategy involves the use of neuronal cultures derived from human induced pluripotent stem cells (iPSCs), which are generated by reprogramming adult somatic cells (e.g., skin or blood) back to a pluripotent state.⁸⁵ Crucially, iPSCs retain the donor's genetic background, allowing the generation of patient-specific neural cells that reflect individual susceptibility to connectome pathology. In parallel, genome engineering techniques, such as CRISPR-Cas9, allow targeted disruption or correction of genes implicated in connectivity disorders, facilitating controlled studies of causal mechanisms in an isogenic background.⁸⁶ Neural stem cell models, grown in two or three dimensions, recapitulate key processes relevant to white-matter abnormalities, including axonal growth, synapse formation, myelination, and neuroinflammation.⁸⁷ Beyond individual cell behaviors, they also support the emergence of network-level properties, enabling investigation of both structural and functional connectivity *in vitro*.^{88,89} When derived from individuals with neurodevelopmental or psychiatric conditions involving known or suspected connectome disruptions—or are engineered to carry risk-associated variants—these models can be experimentally aligned with neuroimaging phenotypes to dissect underlying cellular drivers. While current neural stem cell models do not reproduce the full architectural complexity of endogenous brain circuits, they offer a scalable and human-specific platform for probing the cellular and molecular basis of altered white-matter connectivity and topology observed by neuroimaging.

In the remainder of this section, we consider in more detail how different steps of the connectome-mapping pipeline are affected by specific white-matter pathologies (Figure 3).

The first step is voxel-wise biophysical modeling or signal deconvolution. Many properties of a biophysical model are sensitive to white-matter pathology.^{90,91} The most informative microstructural properties include intra- and extra-axonal volume fractions, fiber orientations, and diffusivities. Unfortunately, most connectome-mapping pipelines do not *directly* utilize these microstructural properties to estimate structural connectivity, although modern pipelines based on microstructure-informed tractography aim to explicitly use intra-axonal volumes to help to quantify connectivity (see [choosing a connectivity measure](#)).^{92–95} Reductions in intra-axonal volume can suggest axonal loss and degeneration (e.g., Alzheimer's disease, Parkinson's disease, Amyotrophic lateral sclerosis), but may also indicate demyelinating pathologies in disorders such as Multiple Sclerosis. In contrast, increases in extra-axonal volume and free water⁸⁰ may suggest the presence of inflammation and edema in conditions such as schizophrenia, traumatic brain injury, and HIV-related neuroinflammation.⁸² Utilizing volume fractions and other microstructural properties to guide structural connectivity quantification may thus enable connectomic differentiation of specific pathologies.

Although most structural connectivity measures do not directly utilize microstructural properties inferred from the diffusion signal, biophysical model features influenced by pathology can nonetheless significantly affect the second step of the connectome-mapping process—tractography. Specifically, the streamline propagation process is directly shaped by estimated fiber orientations, fiber dispersion, fiber orientation coherence between neighboring voxels, as well as diffusivities and confidence metrics—these affect streamline stopping criteria. One key issue in the field is that alterations in these properties are not highly specific to any one pathological process and can be affected by a range of white-matter disorders. For example, tractography can be used to distinguish three types of interactions between white-matter tracts and glioma. Specifically, the propagation of streamline trajectories during the tractography process can either be: (1) displaced, (2) unaffected, or (3) disrupted/halted by the tumor region as defined on imaging. This classification distinguishes three glioma-tract interaction types: displaced (also referred to as deviated), infiltrated, or disrupted (also referred to as destroyed), respectively.⁹⁶ Different classification systems include a fourth edematous type.^{97,98} Streamline tractography remains largely unaffected by the infiltrated type of glioma-tract interaction, whereas the disrupted type causes premature streamline terminations within the tumor volume. Streamline trajectories bend around the glioma and remain contiguous in the displaced type (Figure 4). Interaction type (displaced, infiltrated, disrupted) associates with tumor histopathology, establishing a link between macroscale connectivity and underlying cellular-level pathology.⁹⁶

Importantly, we note that streamline blocking, displacement, or disruption by a tumor should be interpreted as image-based phenomena, with no implication of physical interaction between streamlines and tumor tissue. Throughout this work, references to the effects of pathology on streamline propagation and tractography pertain exclusively to *in silico* behavior within the computational tractography framework, rather than to physical processes occurring in pathological tissue.

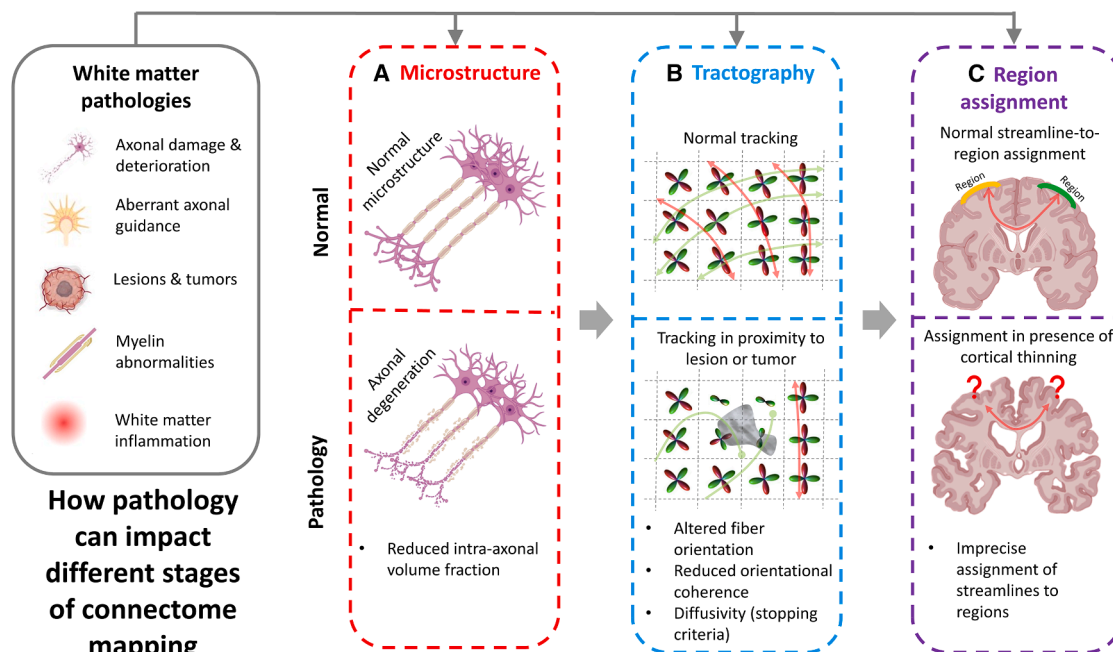


Figure 3. Example effects of white-matter pathology on different stages of connectome mapping

(A) Axonal degeneration, loss, and to some extent myelin-related pathologies can alter intra-axonal volume fractions, in turn affecting structural connectivity derived from microstructure-informed tractography.

(B) Tumors, space-occupying lesions, and other white-matter pathologies can alter streamline propagation. In proximity to the lesion, observe that fiber orientation estimates are perturbed and orientation coherence between neighboring voxels is not evident, hindering or blocking streamline propagation and thus reducing interregional structural connectivity.

(C) The assignment of streamline endpoints to cortical regions can be affected by extensive cortical thinning, altering the estimated connectivity matrix architecture. Some parts of this figure were created with [BioRender.com](https://www.biorender.com).

Tractography-pathology interactions are as poorly characterized for conditions other than glioma. In most cases, work has focused on using tractography to delineate the *spatial extent* of tracts to facilitate neurosurgical planning and targeting,³⁵ rather than our focus here on *connectivity strength quantification* to facilitate connectome mapping. In traumatic brain injuries, shearing forces can distort or deform axonal trajectories, resulting in altered fiber orientations, axonal swelling, and neurofilament loss.⁹⁹ Tissue necrosis and edema caused by stroke can also cause disruption to streamline propagation (i.e., due to Wallerian degeneration) or deformation of streamlines. Disambiguating between these alternative pathological scenarios at the scale of the connectome is currently very challenging (see [future directions and recommendations](#)).

The final step of connectome mapping—assigning streamline endpoints to regions of a brain parcellation¹⁰⁰—can be affected by cortical thinning and gray-matter pathologies. In the presence of extensive cortical atrophy (e.g., chronic schizophrenia, neurodegenerative disorders), the gray-white-matter boundary may be difficult to delineate, and standardized regional parcellation atlases may provide a poor anatomical fit. As such, identifying streamline endpoints at the gray-white-matter boundary may be difficult, and streamline-to-region assignments may be affected by poorly registered parcellation atlases (Figure 3C), altering the connectivity matrix architecture. Although these alterations reflect gray-matter pathological changes, they are generally treated as confounding factors rather than direct indi-

cators of disease because they do not capture connectivity *per se* (see [building a connectivity matrix](#)).

Table 1 summarizes how different white-matter pathologies may impact the connectome mapping-process.

CHOOSING A CONNECTIVITY MEASURE

Workflows and pipelines for connectome building are comprehensively described in prior work,^{9,101,102} and several outstanding primers on tractography-based connectome mapping are available.³ One of the most important decisions to make before building a connectome is choosing a connectivity measure. Ideally, connectivity measures should provide both sensitivity and specificity to distinct white-matter pathologies. Moreover, an ideal measure should scale linearly with pathology severity and extent, enabling quantification of disease progression. These properties are often overlooked by connectome pipeline builders, who place more focus on developing connectivity measures that are: (1) independent of confounds, such as the number of streamlines seeded, fiber bundle length, etc., and (2) scale with some aspect of white-matter microstructure or bundle geometry. While these are also important properties, as outlined below, we contend that they may be less critical for patient-control connectome comparisons. Because of the inherent limitations of tractography, existing connectivity measures do not capture either the directionality of connections or their functional type (e.g., inhibitory versus excitatory). While the absence of directionality information can

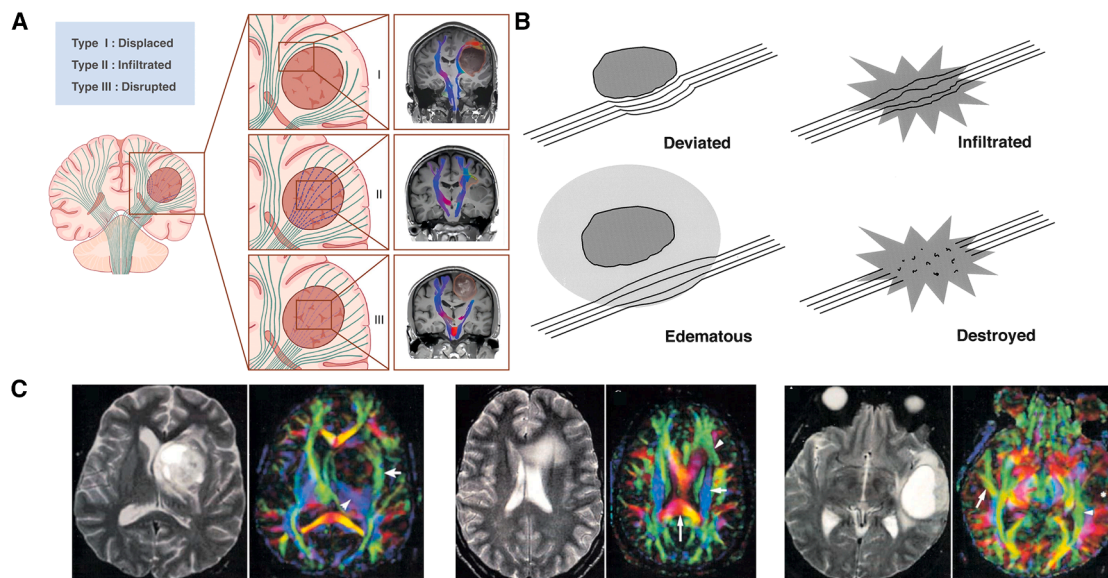


Figure 4. Impact of glioma on streamline propagation

(A) Schematic of three types of glioma-tract interaction patterns, characterized by distinct streamline propagation behaviors, whereby streamlines: (i) are displaced and course around the tumor, (ii) pass through the tumor relatively unaffected, or (iii) are disrupted, resulting in streamline terminations within the tumor mass. Note that streamline blocking, displacement, or disruption by a tumor should be interpreted as image-based phenomena, with no implication of physical interaction between streamlines and tumor tissue. Image reproduced from Hu et al.⁹⁶ with permission.

(B) A similar classification of glioma-tract interaction patterns, but with the inclusion of a fourth edematous pattern. Image reproduced from Jellison et al.⁹⁸ with permission.

(C) Whole-brain tractography in patients with displaced, infiltrated, and disrupted glioma-tract interaction pattern types. The displaced type (left) is in a patient with a pilocytic astrocytoma, and the arrowheads indicate displacement of the corona radiata. The infiltrated type (center) is in a patient with an oligodendroglioma. Arrowheads indicate contiguous streamlines traversing the corona radiata, arcuate fasciculus, and corpus callosum. The disrupted type (bottom row) is in a patient with an astrocytoma, where the asterisk indicates obliteration of fibers in the temporal lobe. Images modified from Witwer et al.⁹⁷

impact connectome inference,¹⁰³ primate tract-tracing studies suggest that most mesoscale connections are bidirectional. For many connections, a bidirectional assumption is likely to be a reasonable approximation.

Current measures fall within five broad categories:

- (1) Streamline counts, also termed the number of streamlines (NOS)^{12,13};
- (2) Filtered streamline counts derived from microstructure-informed tractography, which lead to connectivity measures referred to by various names, including connection density,¹⁰⁴ fiber bundle capacity,¹⁰⁵ and microstructure-informed connectivity^{94,106} as well as others,^{95,107} depending on the type of filter used;
- (3) Tract-averaged and along-tract measures of white-matter microstructure^{108,109};
- (4) Structural covariance and similarity^{110–112}; and,
- (5) AI-based, global, and semi-global tractography measures.^{113–117}

Figure 5 characterizes the first four of these structural connectivity measures when they are used in the presence of pathology.

NOS is the most widely used measure in clinical studies, although microstructure-informed tractography is gaining traction. Tract averages of anisotropy and microstructural measures are less commonly used for connectome building, and structural covariance/similarity is an indirect measure of anatomical con-

nectivity. We do not consider the last category here because it is uncommonly used in clinical connectomics, although global and AI-based connectome mapping provides a promising way forward for the field.¹¹⁸

Streamline counts

To measure connectivity strength between a pair of regions, the total number of tractography-derived streamlines interconnecting the pair can be enumerated. Streamlines are binary objects. Enumerating them to yield streamline counts is quick and conceptually simple. Enumeration can be repeated for all regional pairs comprising a brain parcellation, yielding the connectome's defining structure—a connectivity matrix. Streamline counts are the most widely used measure of structural connectivity in clinical connectomics.

However, streamline counts are increasingly labeled non-quantitative.^{3,43,44,105} This is because streamline counts are not necessarily proportional to the number of axons comprising the underlying fiber bundle or some other microstructural or geometric property of the bundle. Although variation in streamline counts may relate to variation in axonal density and myelination, counts are influenced by numerous confounds, such as unavoidable biases inherent to the tractography algorithm^{3,9} and user-controlled parameters such as the number of streamlines seeded. As such, streamline counts cannot provide an absolute quantification of structural connectivity strength.

Table 1. Putative impacts of specific white-matter pathologies on local microstructural properties, tractography, and structural connectivity estimation

Pathology	Local effects on brain tissue model	Effects on tractography and connectivity
Axonal loss	reduced intra-axonal volume fraction, reduced anisotropy, and blurring/dispersion of fiber orientations	streamline dropout, premature streamline terminations, broken pathways, and reduced connectivity
Demyelination	increased extra-axonal volume fraction, due to expanding extracellular space; relatively preserved fiber orientation—loss of myelin does not immediately distort fiber orientation estimates	relatively preserved fiber tracking and connectivity in early stages, unless secondary axonal loss is present
Tumor or gliosis	disruption of fiber coherence, orientation, and dispersion; tumor growth and glial scarring introduce non-neuronal tissue components (e.g., tumor cells and reactive astrocytes) that physically push fibers aside, bending or angling their orientation away from normal trajectories; loss of tightly packed axons and increased extracellular matrix or tumor cells causes fibers to point in more scattered directions; non-neuronal tissue infiltration alters axonal alignment, reducing the uniformity of fiber directions	streamlines may curve around the tumor, bypass the tumor periphery, or terminate prematurely at tumor margins or glial scar tissue; glioma-tract interaction patterns can be classified based on whether streamlines: (1) are displaced by the tumor, (2) pass through the tumor, or (3) are disrupted by the tumor ⁹⁶
Multiple sclerosis plaques	focal reduction in intra-axonal volume fraction, blurring/dispersion of fiber orientations	focal disruption in streamline tracking, reduced connectivity within specific tracts
Inflammation and edema	increased extracellular water; swelling and immune infiltration physically disturb local fiber orientations; edema pushes axons apart, leading to a broader range of fiber angles and dispersion within a voxel	streamline dropout and premature streamline terminations at the inflammation site, leading to reduced connectivity

Effects on tractography and connectivity refer to processes occurring within the computational tractography framework, rather than physical processes in pathological tissue.

While absolute quantification remains an important consideration for clinical connectomics, compared with healthy connectome mapping, the need for an absolute connectivity measure is less critical, particularly for patient-control connectome comparisons. This is because algorithmic confounds can be held approximately constant when mapping patient and control connectomes using the same tractography algorithm (e.g., deterministic or probabilistic) and parameters (seed density, thresholds) in both groups. Notwithstanding complex interactions between diagnostic status and tractography biases, confounds should cancel out between patients and controls during statistical comparisons, leaving between-group differences that reflect the diagnostic or pathological effect of interest. However, even if all biases do indeed cancel out, it remains difficult to attribute the between-group difference in streamline counts to a specific microstructural cause.

Streamline counts remain entrenched due to their simplicity, being the first measure of their kind and the many studies demonstrating their utility in treatment planning,²⁸ diagnostic prediction and localization of abnormal circuits,²⁴ and topology.¹¹⁹ We contend that streamline counts remain a reasonable connectivity measure for clinical connectome construction, particularly in conditions involving gross tract disruptions. How-

ever, it is important to recognize that newer measures offer greater sensitivity to subtle and/or diffuse pathology, and can provide better quantification of white-matter microstructure, helping disambiguate the biological basis of connectivity changes that streamline counts alone cannot resolve.

Connection density and microstructure-informed tractography

Streamline filtering methods can be applied post hoc to the tractogram before streamline enumeration, with the aim of scaling streamline counts with properties of white-matter microstructure.^{92–95} These methods are also known as microstructure-informed tractography. The basic premise is to either remove streamlines or fit per-streamline multipliers so that the summed weight of streamlines traversing each fiber direction within a voxel scales with properties of the underlying tissue microstructure. For example, under spherical-deconvolution-informed filtering of tractograms (SIFT) methods,^{92,104} streamlines are selectively pruned or down-weighted to achieve a constant proportionality coefficient between the weighted sum of streamlines traversing each fiber direction within a voxel and the corresponding intra-axonal volume estimated from the diffusion signal. In this way, each streamline represents some cross-sectional

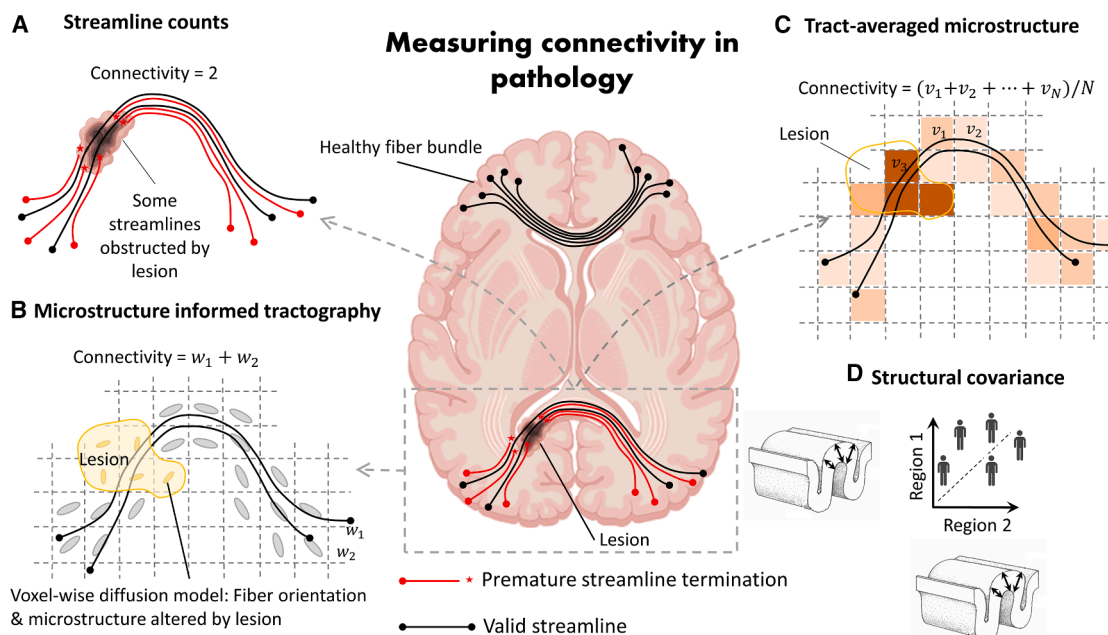


Figure 5. Structural connectivity measures for connectome mapping in disease

Axial brain slice shows tractography streamlines traversing the genu and splenium of the corpus callosum. Some streamlines are obstructed by a circumscribed lesion located in the left splenium, resulting in pathology-based premature streamline terminations (pPST). Each panel shows how different connectivity measures quantify interhemispheric connectivity strength between the left and right occipital lobes in the presence of the lesion.

(A) Stars indicate premature streamline terminations due to lesion obstruction. Terminations reduce the interregional streamline count.

(B) Orientation and size of ellipsoids denote local fiber orientation and axonal density, derived from a voxel-wise biophysical model. Fiber orientation and axonal density is altered by the lesion. The weights (w) fitted to streamlines adjust the contribution of each streamline to better match underlying biological properties, including the impact of the lesion on white-matter microstructure. Connectivity is quantified by the sum of weights across all valid streamlines.

(C) Voxels are colored according to a scalar measure (v) derived from the diffusion signal, such as the apparent diffusion coefficient. Connectivity is measured as the average of the scalar measure across all voxels intersected by a streamline. In turn, lesion-related microstructural abnormalities are reflected in abnormal scalar values along streamlines.

(D) Interindividual correlation in cortical thickness measurements between a pair of cortical regions yields a non-tractography measure of connectivity referred to as structural covariance or structural similarity. While this measure does not directly reflect white-matter connectivity, focal white-matter lesions may weaken covariance by disrupting shared trophic, developmental, or activity-dependent influences between cortical regions. The stick-figure icons are intended to illustrate that structural covariance quantifies variation across individuals, providing a group-level estimate of connectivity. Some parts of this figure were created with [BioRender.com](https://www.biorender.com).

area of axons, and thus the weighted streamline count between a pair of regions yields a connectivity measure reflecting the total intra-axonal cross-sectional area of the axons attributed to the regional pair. This measure of connectivity is also known as connection density or fiber bundle capacity.¹⁰⁵ Other streamline filtering methods, such as convex optimization modeling for microstructure-informed tractography (COMMIT), operate on similar principles but use different biophysical models for the diffusion signal or down-weight streamlines rather than removing them completely.^{94,106}

Microstructure-informed tractography can also refer to tractography algorithms in which the streamline-tracking process is guided by microstructural tissue information, such as axon-informed tractography methods, or AxTract.¹²⁰ These methods use tissue-aware priors to improve the biological plausibility of streamline tracking *per se*, but do not involve any post hoc adjustment of streamline weights.

Application of streamline filters can yield biased estimates of connectivity strength in the presence of white-matter abnormalities, particularly focal pathologies that are circumscribed to a limited portion of a fiber bundle.^{121,122} In particular, the premise

of forcing a global proportionality constant between streamline counts (or per-streamline multipliers) and white-matter microstructure can break down in the presence of focal pathology. This is because proportionality is enforced globally across all white-matter voxels and fiber directions, irrespective of whether local tissue is diseased or healthy. The proportionality constant should ideally covary based on the local tissue state. This effect can, in principle, cause paradoxical patient-control differences in which pathology is detected in normal fiber bundles. However, it is important to note that investigations of these phenomena are limited to hypothetical scenarios, and the extent of potential bias introduced by filtering remains disputed.^{123,124}

In addition to errors introduced by globally enforced proportionality constraints, microstructure-informed filters may also fail when the underlying diffusion models themselves break down in pathological tissue. Streamlines traversing lesioned regions may be disproportionately down-weighted or excluded because the altered diffusion signal violates key model assumptions—such as fixed compartmental diffusivities or stable tissue composition. In such cases, filtering can inadvertently suppress a pathological signal, reducing sensitivity to disease-related

disruption. This highlights a distinct trade-off: while filtering enhances biological interpretability under normative conditions, it risks attenuating meaningful differences in clinical populations when pathological tissue violates the model's core assumptions.

Fortunately, precautions can be taken to alleviate streamline filtering biases when mapping connectomes in patient cohorts. SIFT can be modified by applying a modulation factor to pathological regions,¹²⁵ although this requires a priori knowledge of disease loci, which may not be known in advance for many conditions. Inter-subject normalization may also enable unbiased comparison of SIFT-derived connectivity strengths.¹⁰⁵ In the case of COMMIT, an additional tissue compartment can be introduced into the framework to model focal pathology and yield unbiased connectivity estimates.¹²⁶

Overall, we recommend caution when using streamline filters in the presence of pathology. However, the benefits of streamline filtering outweigh its biases. Recently suggested adjustments to filtering methods, as well as inter-subject normalization, should be used where appropriate to mitigate these biases.

Tract-averaged microstructural measures

Rather than using tractography to directly probe pairwise connectivity strength, an alternative approach is to use it only to define the spatial extent of the fiber tracts interconnecting a pair of regions. Any voxel-wise white-matter microstructural index can then be averaged across the tract's spatial extent to yield a connectivity measure. Metrics derived from the diffusion tensor are commonly used, including fractional anisotropy as well as radial, axial, and mean diffusivity.^{108,127,128} The diffusion tensor is a relatively simple yet robust model of local diffusion within a voxel, estimated from diffusion MRI, that characterizes the principal diffusion direction and its magnitude.

This approach offers two advantages. First, fewer demands are placed on the tractography algorithm. In contrast to connectivity strength quantification, tract delineation is a task for which tractography was originally developed and for which it performs well.¹⁰ Second, microstructural measures derived from the diffusion signal and models such as NODDI can index relatively specific microstructural properties, enabling more biologically informed inference about connectivity abnormalities.

The disadvantage of tract-averaged measures of connectivity is that they provide poor sensitivity to focal pathology. A focal lesion, fiber bundle transection, or some other circumscribed pathology can result in a loss of region-to-region connectivity. The lesion or transection site may encompass a small portion of the overall tract's volume, and thus its contribution to any tract-averaged measure is dwarfed by the contribution from all unaffected segments of the tract. Even if the microstructural measure is sensitive to the abnormality, averaging the measure across all voxels comprising the tract yields an average value that might not be substantially different from the average that would result if the lesion were not present.

To alleviate some of these limitations, tract averages can be replaced with along-tract measures¹⁰⁹ and other descriptive statistics, including the maximum or minimum value of the microstructural measure along the fiber bundle's longitudinal axis, or the maximum or minimum cross-sectional area of the tract mask.^{3,129,130} These extremal statistics can potentially offer

greater sensitivity to focal pathology, although they are particularly susceptible to noise effects and are not commonly used in the field.

Structural similarity and covariance

Structural similarity (also known as structural covariance) provides a correlational measure of connectivity strength that does not rely on tractography or diffusion MRI.^{110,112} The measure is based on the premise that the morphology, size, or microstructural properties of two cortical regions covary across a population of individuals when the two regions are anatomically connected. Morphological covariance may be the result of mutually trophic factors.^{111,131}

Structural covariance is challenging to interpret when morphological cortical abnormalities, such as cortical thinning and volume loss, are present, which may accompany connectome abnormalities. The problem is that disease-related morphological changes can confound structural covariance and be mistaken for genuine connectivity disruptions. For instance, in Alzheimer's disease, widespread cortical atrophy can lead to apparent increases in structural covariance between atrophied regions, not because of altered connectivity, but due to shared disease mechanisms such as tau accumulation and selective vulnerability of default mode network hubs, which may follow anatomical pathways yet do not imply preserved or increased structural coupling. Therefore, it is important to disambiguate the effects of cortical thinning and volume loss from patient-control differences in structural covariance. This disambiguation is not trivial, and related issues may arise.¹³²

Another limitation of structural covariance is that it is a group-level measurement of connectivity strength. Comparing group-level estimates is statistically more challenging than comparing individual connectivity estimates because standard deviations cannot be directly computed. Related approaches, called structural similarity networks, incorporate multiple morphological and microstructural properties to enable estimation of individual-specific connectomes based on covariance or other similarity measures across these properties.^{112,133–135} Further work is needed to elucidate the cellular and molecular basis of structural similarity networks and the extent to which they recapitulate tractography-derived connectivity.

The extent to which structural covariance is sensitive to connectome abnormalities is unclear, although the measure has been used in clinical studies.¹³⁶ It is unlikely that the measure is sensitive to focal pathology (e.g., white-matter infarction), since cortical morphology can remain unaffected in these cases. Due to these shortcomings, we suggest that structural covariance be considered a fallback choice when adequate diffusion MRI data are unavailable for tractography.

TRACTOGRAPHY IN PATHOLOGICAL WHITE MATTER

Once a connectivity measure is chosen, the next step in connectome building is to configure a tractography pipeline and a strategy to assign streamlines to brain regions. Tractography configuration options, user-defined parameters, and algorithmic choices give rise to a complex web of decision points. Here, we do not provide perspectives on all decision points, but rather

focus on the decisions that are most consequential to clinical connectome mapping.

Tractography algorithm

Deterministic or probabilistic fiber tracking? Probabilistic tracking is more prone to reconstructing non-existent connections (false positives) than deterministic methods, whereas deterministic tractography is more susceptible to connection-detection failures (false negatives).^{10,137} As such, deterministic methods yield relatively sparse connectomes with high specificity but low sensitivity, whereas probabilistic connectomes are generally dense, or even fully connected, and connectivity weights furnish evidence for the absence or presence of connections.³

This sensitivity-specificity curse remains a major challenge in the field.¹³⁸ It has been argued that specificity is more important than sensitivity for connectome mapping, supporting the use of deterministic methods.¹³⁹ Moreover, evaluations conducted with numerical connectome phantoms suggest that deterministic tractography may be preferred for connectome mapping.^{140–142} A common fallacy is that deterministic tractography cannot resolve crossing fiber geometries, which is a problem that relates to deficiencies in the local diffusion signal model rather than tractography *per se*. If using deterministic tractography, seeding streamlines uniformly at random rather than at voxel centers adds beneficial stochasticity to the tracking and can facilitate diverse sampling of fiber bundles.

Despite these considerations, probabilistic tractography remains a popular choice for connectome mapping. Probabilistic methods can better characterize the likelihood of connections in the presence of uncertainty. Furthermore, most microstructure-informed tractography approaches have been developed and validated using probabilistic methods; thus, these methods are the preferred choice when a filter is used.

We suggest that deterministic tracking is preferred when: (1) pathology is hypothesized to impact a well-defined tract; or (2) precision and clarity are critical, such as in surgical planning, lesion mapping, or targeted neuromodulation. This is because probabilistic methods can reconstruct fuzzy, dispersed pathways, potentially obscuring the visualization and identification of pathological loci. Probabilistic tractography may provide a better characterization of pathology that is diffuse, multifocal, or affects regions with complex fiber organization, such as in traumatic brain injury, diffuse axonal injury, or neurodevelopmental and neurodegenerative diseases, where uncertainty is high, and sensitivity is prioritized. In the future, tractography accuracy can be improved by leveraging information beyond local orientation estimates.¹⁴³ This could include information about pathological loci, priors based on neuroanatomy,¹⁴⁴ and other modalities.¹²⁰

Streamline propagation to probe pathology

Tractography streamlines are seeded at points in the white-matter volume or on the gray-white interface. After initialization, a streamline is propagated step-by-step until both of its endpoints reach a gray-matter region. But some streamlines will terminate prematurely, before reaching gray matter. Premature streamline termination may be an indicator of white-matter pathology. For

example, streamlines can be obstructed by a lesion (Figure 4), terminate or veer off course due to disease-related torsional force on the estimated fiber orientations, or terminate within pathological white-matter voxels at which the minimum anisotropy threshold is not met. The number of prematurely terminating streamlines can index the extent of the pathology's impact on connectivity.

The problem is that pathology is not the only cause of premature streamline terminations—errors in tractography, partial volume effects, misalignment and registration errors, and noisy data also cause terminations.¹⁴⁵ These errors are characterized by unexpected terminations in cerebrospinal fluid, exceedance of streamline curvature thresholds, and other anatomically implausible trajectories. As such, two types of premature streamline terminations can be distinguished: *pathology-based premature streamline terminations* (pPST) and *erroneous premature streamline terminations* (ePST). Streamlines that do not terminate prematurely are called *valid streamlines*. Both endpoints of a valid streamline must reside in gray matter. Only valid streamlines can contribute to streamline counts and connectivity matrix construction.

Conventional connectome-mapping pipelines leverage pPST events to reduce region-to-region streamline counts to capture the impact of pathology on connectivity. Under conventional pipelines, a fixed number of streamlines is initialized for each patient and healthy comparison individual. Therefore, the number of ePST events should not significantly differ between the patient and control groups, given that the same tractography algorithm is used and the same number of streamlines is seeded. Under this assumption, on average, patient-control differences in streamline counts can be specifically attributed to pPST events. In some cases, assuming equal ePST rates between groups may be problematic due to amplified confounds in one group (e.g., increased head motion or poorer registration in patients compared with controls).

The streamline propagation process itself serves as a test probe for pathology along a fiber bundle. If the probe is obstructed, veers off course, loops back on itself, etc., either a pPST or ePST event is detected. Unfortunately, it is not possible to distinguish between pPST and ePST events. However, the number of ePST events should be comparable between patients and controls, meaning that a greater number of terminations (equivalent to lower streamline counts) in the patient group provides a marker of pathology-specific disrupted connectivity. Under this rationale, streamline filters are not used because the streamline count itself directly encodes the impact of white-matter pathology on connectivity. This approach was used to map connectome pathology before the establishment of the first streamline filters.^{16,24} Nevertheless, we reiterate that differences in streamline counts between patients and controls may not solely reflect pathological effects, but could also be influenced by methodological confounds.

When using conventional connectome-mapping pipelines, it is critical to ensure that premature streamline terminations do not trigger seeding of new streamlines. The default behavior of some tractography software packages is to weed out premature terminations from the tractogram and initialize a new streamline for each streamline weeded out. This ensures that the total

Table 2. Characteristics of two broad frameworks for mapping clinical connectomes

Property	Connectome-mapping framework	
	Conventional	Microstructure-informed
Streamline filtering?	not required	yes
Cull prematurely terminating streamlines?	no	yes
Deterministic or probabilistic tracking?	both	probabilistic preferred
Number of streamlines seeded?	fixed	variable
How is connectivity measured?	streamline counts	microstructure, biophysical model
How is pathology indexed?	premature streamline terminations	change in microstructural properties
Computational complexity	low	modest

number of valid streamlines is fixed at a user-defined number. For healthy connectome mapping, or when a streamline filter is subsequently applied to the tractogram, this behavior can be desirable. However, for patient-control connectome comparisons performed without a streamline filter, prematurely terminating streamlines should not be replaced with new ones.

It is important to consider why this is the case. Premature terminations will be more frequent in the patient group due to pPST events. If new streamlines are seeded, healthy connections may be oversampled in patients relative to the control group. Although streamline seeding is uniform, if by chance the new seed samples a pathological bundle again, a further pPST event is likely, resulting in seeding another new streamline. This re-seeding process continues until a healthy bundle is eventually sampled and a valid streamline is generated, artificially inflating streamline counts in preserved tracts in the patient group compared with the control group. Inter-subject normalizations can be devised to alleviate this bias, although not replacing premature terminations with new streamlines is the simplest solution.

The conventional connectome-mapping pipeline described above inherits shortcomings of the streamline count as a measure of connectivity strength. Most notably, the impact of pathology on connectivity cannot be quantified in an absolute sense. It can only be quantified through relative comparison with a healthy control group or other individuals. The conventional approach also assumes that ePST frequencies do not differ between patients and controls.

Microstructure-informed pipelines incorporate a streamline filter to address many of these shortcomings. They comprise four key steps: (1) whole-brain tractography; (2) weeding out prematurely terminating streamlines and replacing them with new ones¹⁴⁵; (3) applying a streamline filter to the resulting tractogram; and (4) connectivity matrix construction. Filtering ensures that connectivity measurements capture some property of underlying white-matter microstructure, such as the total intra-axonal cross-sectional area in the case of SIFT¹⁰⁴ or the axonal volume fraction for COMMIT.⁹⁴

In patient-control connectome comparisons, pPST events will trigger the initialization of new streamlines in the patient group and potentially lead to the same oversampling bias described above. It is then the responsibility of the streamline filter to correct for these and any other tractography sampling biases. As discussed in an earlier section, precautions may be needed when applying streamline filters in the presence of pathology.

The advantage of microstructure-informed connectome mapping is that it enables absolute quantification of connectivity strength over unadjusted streamline counts.

In summary, clinical connectome-mapping pipelines fall within two frameworks: (1) conventional, and (2) microstructure-informed. Conventional pipelines embrace pPST events and use their impact on streamline counts to directly probe pathology. If more probes can successfully propagate to both endpoints of a fiber bundle in the control group compared with patients, a disease-specific disruption in connectivity can be inferred. Connectome-mapping pipelines in this category do not require a streamlines filter. In contrast, microstructure-informed pipelines rely on streamline filters to ensure that streamlines are endowed with per-streamline multipliers that reflect underlying white-matter microstructure. In this way, the impact of pathology on connectivity strength is captured via a biophysical model. Table 2 summarizes the salient features that differ between the two broad frameworks.

While both frameworks are well-suited to mapping clinical connectomes, microstructure-informed approaches offer a more quantitative characterization of connectivity. Nevertheless, conventional methods have been more extensively tested in clinical settings to date²⁹ (see also Figure 1), whereas microstructure-informed connectome mapping has focused on the healthy brain and may benefit from further validation in the presence of white-matter pathology.

BUILDING A CONNECTIVITY MATRIX

The final step in the connectome building process is assigning streamline endpoints to regions of a brain parcellation atlas.^{100,146} The result of this step is a connectivity matrix. The choice of the brain parcellation atlas is the first decision to make.¹⁴⁷ Whole-brain atlases vary in granularity, spanning from around a hundred cortical and subcortical regions to tens of thousands of parcels defined by cortical surface vertices. The principles governing atlas selection in clinical connectomics are similar to those used in healthy connectome mapping. These include atlas reliability and anatomical validity, computational feasibility, and application-specific considerations.¹⁴⁸ For example, investigating large-scale network organization can benefit from a relatively coarse parcellation, whereas localizing a pathological hotspot to facilitate precision therapeutic targeting mandates the use of high-resolution atlases.

An underappreciated consideration is that connectome abnormalities are often accompanied by cortical thinning and atrophy.^{25,149,150} This presents several challenges. Standardized cortical surface meshes and parcellation atlases may provide a poor fit for patients with extensive cortical atrophy or lesions.¹⁵¹ Even delineating the interface between gray and white matter can be challenging, and specialized segmentation tools may be required.¹⁵² As atrophy progresses, sulci and gyri can become shallow, distorted, and fuzzy, making it difficult to identify anatomical landmarks that define a parcellation. Furthermore, separating the cortex from the underlying white matter can be challenging in atrophied regions in which there is minimal separation, leading to inaccurate tissue boundaries.

Careful inspection of cortical surface reconstructions is essential before proceeding to connectivity matrix construction. Tissue segmentation tools sometimes include an option for a pathological tissue class to explicitly model the effects of well-defined lesions. But this is not helpful for the diffuse and subtle changes characteristic of psychiatric disorders. Moreover, segmentation in the presence of brain tumors and lesions requires specialized segmentation tools. Pathology-related misclassification of tissue types and blurring of cortical boundaries can introduce biases into the connectivity matrix. Connectome changes may be a compensatory outcome of cortical thinning, or alternatively, the result of brain parcellation inaccuracies caused by cortical atrophy.¹⁵⁰

To minimize the conflating effects of gray-matter pathology on connectome mapping, robust and relatively coarse parcellation strategies that do not rely on fine-grained, individual-specific tissue segmentations can be beneficial. For example, volumetric parcellations comprising a few hundred regions may be a good starting point when cortical atrophy is extensive, diffuse, and not localized to distinct lesions. This contradicts the direction of advances in healthy connectome mapping, in which high-resolution connectome mapping^{153,154} and cortical surface parcellations are increasingly used, due to their precision and better alignment across individuals.^{155,156} It can be challenging to capitalize on this precision and superior alignment when extensive gray-matter pathology is present; hence, coarser, volumetric approaches may be sufficient. That said, it is important to acknowledge that evaluations in schizophrenia suggest that surface-based representations may remain beneficial, even in the presence of cortical atrophy.¹⁵⁷ The choice between volumetric and surface-based parcellation is not clear-cut and should be guided by the extent of atrophy (i.e., volumetric may be preferred in severe atrophy), quality of the reconstructed cortical surface, and requirements of any subsequent analyses.

Once a brain parcellation is selected, streamline endpoints must be assigned to parcels to populate a connectivity matrix for each individual. Endpoint assignment is a conceptually simple process, but several different implementations can be found. A source of ambiguity stems from the fact that streamlines can continue to propagate within gray matter and intersect multiple parcels before satisfying the termination criteria. While this propagation within gray matter may reflect erroneous reconstruction, the question remains: Which parcel should the streamline be assigned to in this situation? Some approaches handle this ambiguity by halting streamline propagation at the interface between

gray and white matter. This means that streamlines are effectively assigned to the first gray-matter parcel that they reach. While this approach mitigates spurious multi-parcel assignments in healthy tissue, it may produce inaccurate endpoints when the interface between gray and white matter might not necessarily align with the margins of the chosen parcellation atlas. In this case, a local neighborhood search can be used to match the streamline's endpoint on the interface with the closest voxel or cortical surface vertex of the parcellation.¹⁰⁰ These approaches require accurate delineation of the interface between gray and white matter, which can be challenging when extensive cortical atrophy is present.

Other assignment strategies include adding connections between all regions intersected by a streamline, meaning that a single streamline can contribute to multiple entries in the connectivity matrix. The impact of streamline assignment strategies on clinical connectomics is not well studied, so it is difficult to provide specific recommendations. When cortical atrophy is extensive, it is important to recognize that the interface between gray and white matter may be fuzzy, difficult to delineate accurately, and that cortical parcellations may spill over into white matter due to poor spatial alignment. In these cases, some degree of streamline assignment inaccuracy is inevitable regardless of the assignment strategy, including biases toward gyral crowns and tracking through superficial white matter.⁹ Connectome spatial smoothing¹⁵⁸ can potentially reduce these inaccuracies when mapping connectomes at high resolution; otherwise, the use of a relatively coarse parcellation may smooth out some inaccuracies.

FUTURE DIRECTIONS AND RECOMMENDATIONS

Connectomics with diffusion MRI and tractography presents considerable clinical promise. Real-world clinical translation is already apparent, from identification of symptom-specific circuit targets for deep brain stimulation²⁹ to neurosurgical planning.³⁴ Yet, the extent of clinical integration has been limited, and the full potential of clinical connectomics remains untapped. Moreover, despite extensive advances in connectome mapping over the last decade, clinical uptake of the state-of-the-art in connectomics and diffusion MRI tractography remains modest.

Understanding the key reasons for the limited clinical translation of advances in human connectomics is important. Connectomics faces similar barriers as other technological advances in the last few decades. For example, artificial intelligence in healthcare, genomic medicine, and ultra-high-field MRI face barriers associated with a lack of methodological standardization, regulatory hurdles, and uncertainty about clinical utility.¹⁵⁹ In addition to these well-known barriers, clinical connectomics must overcome unique, field-specific challenges.

What are the unique barriers facing clinical connectomics? Foremost, most new tools and methodologies for connectome mapping are established, tested, and validated using healthy participants. As such, a method's suitability for mapping diseased connectomes can often remain untested until it is first adopted clinically. Connectome tool developers should ideally undertake testing of their methods and models using clinical participants or synthetic data featuring simulated white-matter

pathology. An advantage of synthetic test data is that the ground truth is known, enabling quantitative evaluation of the sensitivity and specificity of a connectivity measurement. However, using biophysical models to generate synthetic data characterizing white-matter pathology is challenging due to limited knowledge about which properties of the model should be adjusted *in silico* to model the pathology's impact. A vast number of biophysical model properties can be adjusted, ranging from the loci of pathological voxels to compartment fractions, estimated fiber directions, orientation dispersion, diffusivities, and other model-specific attributes.⁹⁰ Alternatively, Monte Carlo simulations can be used to generate diffusion signals for morphological models of pathological neurite morphology.¹⁶⁰ Connectomes can then be mapped for the simulated diffusion data to elucidate the impact of microstructure-defined pathology on interregional connectivity.

Validation using synthetic models of pathology is only the first step in the process of clinical translation. If a connectome-mapping pipeline is found to yield reliable and reproducible quantification of a connectome abnormality *in silico*, the next step is to undertake biological and clinical validation to assess whether the quantification is clinically relevant for disease monitoring, outcome prediction, diagnosis, or screening. Many current connectome tools and methodologies have not undergone this kind of validation. Given the substantial effort involved in conducting biological and clinical validation, it is unrealistic to require such validation to be fully completed prior to releasing a tool to the community—especially when the tool is also designed for non-clinical use. Nevertheless, the sensitivity and specificity of a connectome pipeline to a given disease should not be assumed, and must be critically assessed by those engaged in translational work.

This brings us to a second key barrier and knowledge gap. Current interregional connectivity measures provide relatively limited specificity to distinguish between different pathology types. A patient-control difference in the streamline count can result from fewer interconnecting axons (e.g., axonal transection or loss), axonal demyelination, swelling or beading *without* axonal loss, deformation of axonal trajectories, or multiple overlapping pathological changes. This lack of pathological specificity limits the insight that clinical connectomics can provide. Pathological loci can be accurately localized to specific connections and circuits, but beyond anatomical localization, the pathobiological underpinnings of patient-control streamline count differences remain ambiguous. Future research efforts should thus focus on establishing connectivity measures that provide adequate specificity to differentiate microstructural changes.

Of course, this is easier said than done. Microstructure-informed tractography is an established and promising way forward to gain insight into the biological basis of connectome abnormalities. Under these methods, patient-control connectivity differences can be attributed with some confidence to a specific microstructural white-matter property, such as intra-axonal cross-sectional area.¹⁰⁵ Differences in intra-axonal volume/area fractions and neurite density are more likely to indicate damage or loss of axons, including axonal transection, Wallerian degeneration, and diffuse axonal injury, whereas demyelinating pathologies associated with Multiple Sclerosis are likely to

have a smaller impact on volume fractions, at least in early disease stages.¹⁶¹ Microstructure-informed tractography can thus enable some degree of pathological specificity, although it is important to remember that this specificity is based on model-derived estimates rather than direct histological comparison.

More work is needed to elucidate the histopathological correlates of the biophysical diffusion models used to devise per-streamline multipliers in microstructure-informed tractography. Improving and proving the pathological specificity of current biophysical models is very challenging due to fundamental concerns about the inadequacy of current biophysical models, degeneracy in model parameter estimates (i.e., many pathologies can lead to the same diffusion signal), and the difficulty in distinguishing acquisition/modeling variability from biological variability.^{91,162} In the future, biophysical models such as NODDI⁸¹, CHARMED,¹⁶³ or SANDI¹⁶⁴ could be leveraged to inform tractography in the same way as signal deconvolution and ball-and-stick models are currently used in microstructure-informed tractography. Models such as NODDI can provide greater specificity because they model biologically specific compartments,⁹⁰ and some degree of histopathological validation has already been undertaken.¹⁶⁵ For example, under NODDI, a decrease in the neurite dispersion index is consistent with axonal degeneration (e.g., Alzheimer's disease), whereas an increase can suggest inflammation and cellular swelling due to gliosis (e.g., traumatic brain injury). How to integrate models such as NODDI into connectome-mapping pipelines and tractography remains an open question and will likely require fundamental advances in tractography and biophysical modeling of the diffusion signal.

Although the effects of pathology on biophysical modeling of diffusion signals are the most studied, they represent just one aspect of the connectome-mapping pipeline capable of capturing pathological changes. Disease can also affect the tractography process itself (i.e., streamline tracking) and streamline assignments, manifesting changes in interregional connectivity strength and network organization. For example, white-matter lesions may hinder or block streamline propagation, whereas cortical malformations and neurodegeneration may affect streamline terminations and assignments at cortical regions, leading to apparent hypo-connectivity. Alternatively, edema or inflammation can introduce free water and isotropic diffusion, which reduces directional coherence across neighboring voxels and increases the likelihood of altered streamline reconstruction. The effects of pathology on streamline tracking are poorly understood. Multimodal imaging—e.g., combining diffusion MRI with myelin-sensitive contrasts or PET imaging of putative inflammation, synaptic density, and neurotransmitter systems—alongside network-scale histopathological validation, is needed to address this gap.

A third obstacle to clinical adoption stems from the substantial technical challenges that translational researchers encounter when applying connectome-mapping tools and models. Compared with mapping functional connectivity using resting-state fMRI, mapping structural connectivity with diffusion MRI and tractography is arguably more technically complex. Moreover, clinical researchers may not fully appreciate the limitations of connectome mapping in individuals with white-matter

pathology and overlook that many connectome-mapping pipelines are validated using non-clinical participants. Strengthening collaborations between tool developers and clinical researchers is key to overcoming this barrier. Training programs focused on proper tool use and connectome analysis could also offer significant value to clinical researchers. The past decade has seen remarkable progress in connectome-mapping methods, but without widespread adoption by neuroscientists and clinicians, these advances may have little practical effect. Legacy models such as the diffusion tensor remain commonly used in clinical studies, and while the tensor is valid in certain contexts, more informative and specific alternatives are available. It is thus important for the field to address the barriers hampering the uptake of the state-of-the-art in connectome mapping. Recent glioma studies showcase rigorous clinical application of advanced quantitative tractography methods.^{96,166}

In conclusion, we propose four recommendations to encourage best practices in clinical connectomics and to support the field's progression:

- **Patient-control comparison of network measures:** A vast array of network measures can be statistically compared between patient and control connectomes. Exhaustively testing a comprehensive list of topological measures for patient-control differences is discouraged due to the inter-relatedness of graph-theoretic measures, exacerbation of the multiple comparisons problem, and the difficulty in relating patient-control differences in abstract network properties to concrete pathological processes. In graph-theoretic investigations, we advocate selecting a focused set of network metrics, chosen based on relevance to the disease context and designed to test a specific hypothesis. Additionally, network connection density should be matched or controlled for between groups to prevent density effects from confounding patient-control comparisons in higher-order network measures.¹⁶⁷
- **Connectome-mapping pipelines:** Most connectome-mapping pipelines are tested and validated using non-clinical participants. The extent to which a pipeline can reliably and accurately characterize connectome abnormalities in clinical cohorts is often underexplored. We recommend that connectome tool developers undertake basic testing of new tools using clinical participants or synthetic data featuring simulated white-matter pathology, if clinical applications of their tool are foreseen. We further recommend that translational researchers recognize that many connectome-mapping tools may not yet be validated in pathological white-matter conditions.
- **Pathological specificity is needed:** While connectome abnormalities can be precisely mapped to specific connections, circuits, networks, or topological features, uncovering their underlying pathobiological mechanisms remains a challenge. We recommend that future work should focus on developing structural connectivity measures that provide sufficient specificity to distinguish between distinct pathological processes. Microstructure-informed tractography is an important development toward this goal, although the extent to which current methods can distin-

guish between axonal loss, demyelination, gliosis, etc., remains to be established.

- **It's not about right or wrong; it's about what model best fits the research question:** There is no absolute standard in clinical connectome mapping. The choice of connectome-mapping pipeline and model suitability depends on the disease being studied, the research question, data quality, and many other factors. We urge researchers to critically evaluate advice suggesting that particular models and connectome-mapping pipelines are universally appropriate or inappropriate. Despite their limitations, deterministic tractography and the diffusion tensor are not inherently inappropriate models, and they have proven useful in characterizing connectome abnormalities,²⁴ guiding treatment optimization,²⁸ and tools to support retrospective clinical analyses, e.g., Lead-DBS (deep brain stimulation).²⁷ Given the current gaps in our understanding of disease-related changes in structural connectivity, we propose that strict standardization of clinical connectome mapping is not yet warranted.

Actioning these recommendations will accelerate development and enable broader translational applications, including optimization of brain stimulation targets, neurosurgical planning, and connectome-based prognostic biomarker development. While important barriers must still be addressed, there is strong reason to be optimistic about the future of clinical connectomics.

ACKNOWLEDGMENTS

At the time this work was undertaken, A.Z. was supported by the ARC Future Fellowship (FT220100091) and the Rebecca L. Cooper Fellowship. M.A.D.B. was supported by the Val and Al Rosenstraus Fellowship and the NHMRC Emerging Leadership Fellowship.

The authors are grateful to the four anonymous reviewers for providing comprehensive and valuable feedback on the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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